

MODULE 2: CRYPTOGRAPHY

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Cryptography

Cryptography is the science and practice of securing information and communications through the use of codes, protecting it from unauthorized access or alteration. At its core, it transforms human-readable data (**plaintext**) into an unreadable, scrambled format (**ciphertext**) using mathematical algorithms and a secret key. Only someone with the correct key can reverse the process to read the original message.

- **Plaintext:** Original message
- **Ciphertext:** Encrypted message
- **Encryption:** Plaintext → Ciphertext
- **Decryption:** Ciphertext → Plaintext
- **Key:** Secret value used in encryption/decryption

Key space = total number of possible keys. Example: 56-bit key → 2^{56} possibilities. Larger key space → More secure against brute force.

Transformation of plaintext into ciphertext

1. Substitution Techniques

Substitution ciphers replace plaintext characters with other characters, numbers, or symbols while keeping their positions unchanged. The actual letters are substituted according to a specific rule or key. These techniques are among the oldest methods used in classical cryptography. Substitution techniques are categorized based on:

- Number of alphabets used
- Number of letters encrypted at a time

The major substitution techniques are:

1. Caesar/Shift Cipher
2. Monoalphabetic Cipher

3. Playfair Cipher
4. Hill Cipher

5. Vigenère Cipher

1. **Caesar Cipher (Shift Cipher):** The Caesar Cipher is one of the oldest and simplest encryption techniques. It was used by the Roman emperor Julius Caesar for secure military communication. In this cipher, each letter in the plaintext is shifted by a fixed number of positions in the alphabet.

Encryption Formula: $C = (P + K) \bmod 26$

C= Cipher letter

P=Plaintext letter

K=Shift Key

Example

Plaintext Ciphertext(shift=3)

HELLO KHOOR

Advantages

- Very simple to implement
- Fast encryption and decryption
- Useful for learning cryptography basics

Weaknesses

- Extremely weak security
- Only 25 possible keys exist
- Easily broken using brute force attack

2. **Monoalphabetic Cipher (Arbitrary Substitution Cipher):** In a Monoalphabetic Cipher, each plaintext letter maps to a unique ciphertext letter. Unlike Caesar Cipher, the substitution is completely random rather than using a fixed shift. A single substitution rule is used throughout the message.

How It Works

Example mapping:

Plain Letter	Cipher Letter
A	Q
B	X
C	M
D	P
...	...

The key is a complete permutation of the alphabet.

Example

Plaintext: CAT

Using mapping:

$C \rightarrow M$

$A \rightarrow Q$

$T \rightarrow L$

Ciphertext: MQL

Advantages

- More secure than Caesar Cipher
- Large number of possible keys
- Harder to guess manually

analysis because letter frequencies remain unchanged. For example:

- E is the most common English letter
- T and A are also frequent

Attackers can analyze ciphertext frequency to guess mappings.

Weaknesses

Although the key space is very large: $26! \approx 4 \times 10^{26}$. It is still vulnerable to frequency

3. **Playfair Cipher (Polygraphic Substitution Cipher):** The Playfair Cipher encrypts pairs of letters (digraphs) instead of single letters. It was the first practical digraph substitution cipher and

provides better security than monoalphabetic ciphers. For each occurrence of a letter in plaintext, while converting it to ciphertext, each time it will be different letter.

How It Works: A 5×5 matrix is generated using a keyword. (26 letters, 5×5 and I/J together in 1 cell)

Example

Keyword: MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Encryption Rules

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | $\downarrow$ | Wrap around.
4. Same row | $\rightarrow$ | Wrap around.
5. Rectangle | $\bowtie$ | Swap

Rule 1: Same Row: go right

Rule 2: Same Column: go down

Rule 3: Rectangle Rule: If letters form a rectangle: Replace each letter with the letter in the same row but opposite corner column.

Example

Plaintext: attack

Digrams: at ta ck

Plaintext: neso academy

Digrams: ne so ac ad em yx

Plaintext: balloon

Digrams: ba ll oo n

Digrams: ba lx lo on

Example:

1) **Plaintext:** attack

Digrams	at	ta	ck
Ciphertext	rs	sr	Dk

(here we used rectangle rule)

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Ciphertext: rssrdk

② plaintext: mosque

mo	sq	ue
on	ts	ml

① mo in same row, $\therefore$ go right for both
 $m \rightarrow o$, $o \rightarrow n$

② Same for sq $\rightarrow$ ts

③ ue in same row, go down for both.
 $u \rightarrow m$ $e \rightarrow l$

Ciphertext: ontsml

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Question: Encrypt the message "Hide the gold under the carpet" using the keyword "Neso Academy"

N	E	S	O	A	H	I	D	E	T	H	E	G	O	L	D	U	N	D	E	R	T	H	E	C	A	R	P	E	T	X
C	D	M	Y	B	I	K	G	D	Q	K	D	P	N	R	C	V	E	C	O	P	Q	K	N	O	T	V	D	R	Z	
F	G	H	I	J	K																									
L	P	Q	R	T																										
U	V	W	X	Z																										

Advantages

- Stronger than simple substitution ciphers
- Hides single-letter frequency patterns
- More secure than Caesar and Monoalphabetic ciphers

Weaknesses

- Complex implementation
- Still vulnerable to cryptanalysis
- Repeated digraph patterns may reveal information

4. Hill Cipher (Polygraphic Matrix Cipher)

The Hill Cipher uses linear algebra and matrix multiplication for encryption. It is a multi-letter cipher that is it encrypts groups of letters simultaneously rather than individual letters (digraph, trigraph or polygraph). It is mathematically stronger than earlier classical ciphers.

Encryption Formula $C = K \times P \pmod{26}$

Decryption Formula $P = K^{-1} \times C \pmod{26}$

$$K^{-1} = (|K|^{-1}) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$|K|^{-1}$ is the multiplicative inverse of determinant modulo 26.

Example

Using Hill Cipher Encrypt & Decrypt the Plaintext "HELP" using $K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$

Ans) $H = 7$ $E = 4$ $L = 11$ $P = 15$

$P_1 = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$ $P_2 = \begin{pmatrix} 11 \\ 15 \end{pmatrix}$

$$[C_i \ C_j] = (K \cdot P) \pmod{26}$$

$$\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 7 \\ 4 \end{pmatrix} \pmod{26} = \begin{pmatrix} 33 \\ 34 \end{pmatrix} \pmod{26} = \begin{pmatrix} 7 \\ 8 \end{pmatrix} = \begin{pmatrix} H \\ I \end{pmatrix}$$

$$\begin{pmatrix} C_3 \\ C_4 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 11 \\ 15 \end{pmatrix} \pmod{26} = \begin{pmatrix} 78 \\ 97 \end{pmatrix} \pmod{26} = \begin{pmatrix} 0 \\ 19 \end{pmatrix} = \begin{pmatrix} A \\ T \end{pmatrix}$$

$\therefore$ Ciphertext = HIAT

Decryption

$$K^{-1} = 9^{-1} \begin{pmatrix} 5 & -3 \\ -2 & 3 \end{pmatrix} \mod 26 \left[\begin{array}{l} 9^{-1} \mod 26 = 3 \\ \text{or } (7 \times 3) \mod 26 = 1 \\ \hline -x \mod p = (-x + p) \end{array} \right]$$

$$= 3 \begin{pmatrix} 5 & 23 \\ 24 & 3 \end{pmatrix} \mod 26$$

$$= \begin{pmatrix} 15 & 69 \\ 72 & 9 \end{pmatrix} \mod 26 = \begin{pmatrix} 15 & 17 \\ 20 & 9 \end{pmatrix}$$

Ciphertext = H I A T

$$C_1 = \begin{pmatrix} H \\ I \end{pmatrix} = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$$

$$C_2 = \begin{pmatrix} A \\ T \end{pmatrix} = \begin{pmatrix} 0 \\ 19 \end{pmatrix}$$

$$\therefore P_1 = \begin{pmatrix} 15 & 17 \\ 20 & 9 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \end{pmatrix} = \begin{pmatrix} 241 \\ 212 \end{pmatrix} \mod 26 = \begin{pmatrix} 7 \\ 4 \end{pmatrix} = \begin{pmatrix} H \\ E \end{pmatrix}$$

$$P_2 = \begin{pmatrix} 15 & 17 \\ 20 & 9 \end{pmatrix} \begin{pmatrix} 0 \\ 19 \end{pmatrix} \mod 26 = \begin{pmatrix} 323 \\ 171 \end{pmatrix} \mod 26 = \begin{pmatrix} 11 \\ 15 \end{pmatrix} = \begin{pmatrix} L \\ P \end{pmatrix}$$

$$\therefore \text{Plaintext} = \begin{pmatrix} H & E & L & P \end{pmatrix}$$

Advantages

- Stronger mathematical security
- Encrypts multiple letters together
- Provides good diffusion

Weaknesses

- Complex calculations
- Requires matrix operations
- Vulnerable to known-plaintext attacks if enough plaintext-ciphertext pairs are known

5. Polyalphabetic Cipher: A Polyalphabetic Cipher uses multiple substitution alphabets instead of a single alphabet. This makes it more secure than monoalphabetic ciphers because the same plaintext letter can be encrypted into different ciphertext letters at different positions. The most famous polyalphabetic cipher is the **Vigenère Cipher**.

Vigenère Cipher: The Vigenère Cipher encrypts text using a keyword. Each letter of the keyword determines how much the plaintext letter is shifted. The Vigenère Cipher uses multiple Caesar ciphers based on a keyword. It is called a polyalphabetic cipher because different alphabets are used during encryption. This makes it more secure than simple substitution methods.

How It Works: A keyword is repeated until it matches the plaintext length. Each keyword letter determines the shift amount.

Encryption Formula	$C_i = (P_i + K_i) \mod 26$
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Decryption Formula	$P_i = (C_i - K_i) \mod 26$
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Example

Encrypt & decrypt. PT: HELLO Key: KEY

Encryption

PT	H	E	L	L	O	→	7	4	11	11	14
Key	K	E	Y	K	E	→	10	4	24	10	4
CT	R	I	J	V	S	←	17	8	9	21	18

$CT = (K + P) \bmod 26$ [CT: RIJSVS]

Decryption

CT	17	8	9	21	18
Key	10	4	24	10	4
PT	7	4	-15	11	14

= 11
[PT: HELLO]

Advantages

- More secure than Caesar Cipher
- Reduces effectiveness of frequency analysis
- Uses multiple substitution alphabets

Weaknesses

- If key length is discovered, cipher becomes weak
- Vulnerable to Kasiski examination
- Repeating keywords reduce security

2. Transposition Techniques

Transposition ciphers do not replace characters. Instead, they apply some sort of permutations on the plaintext letters that is rearrange the positions of characters according to a specific rule. The letters remain unchanged, but their order changes.

- 1. Rail Fence Cipher (Zig-Zag Cipher):** The Rail Fence Cipher is the simplest transposition cipher. Characters are written diagonally across multiple rows called rails and then read row by row.

How It Works

Example: Encrypting DEFENDTHEWALL using 3 rails:

Plaintext

D . . . N . . . W . . .	→ DNNW
. E . E . D . H . A . L	→ EEDHAL
. . F . . . T . . . L .	→ FTL

Ciphertext: DNNWEEDHALFTL

Advantages

- Simple and fast
- Easy implementation
- Good for basic encryption

Weaknesses

- Weak security
- Easily broken by pattern recognition
- Not suitable for modern security

Rail Fence Technique

Plaintext : neso academy is the best.

Depth : 2

n	s	a	a	e	y	s	h	b	s	
	e	o	c	d	m	i	t	e	e	t

Ciphertext : NSAAEYSHBS

2. Row Column Transposition Cipher: The Row Column Transposition Cipher rearranges plaintext using columns and a keyword. Plaintext is written row by row into a table, and columns are read according to the alphabetical order of the keyword. Example:

Plaintext: ATTACKATDAWN

Keyword: SECRET

Assign column order alphabetically: S(5) E(2)

C(1) R(4) E(3) T(6)

Arrange plaintext in rows:

S (5)	E (2)	C (1)	R (4)	E (3)	T (6)
A	T	T	A	C	K
A	T	D	A	W	N

Columns are read according to numerical order.

Ciphertext: TDTTCWAAAANK

Advantages

- Better security than Rail Fence Cipher
- Harder to detect patterns
- More complex arrangement

Weaknesses

- Still vulnerable to cryptanalysis
- Long messages may reveal patterns
- Not secure against modern attacks

Basis	Substitution Techniques	Transposition Techniques
Definition	Replaces plaintext characters with other characters/symbols	Rearranges the positions of plaintext characters
Main Idea	Character substitution	Character permutation
Security Basis	Hides letter identity	Hides letter position
Encryption Method	Each letter mapped to another letter	Letters shuffled according to a pattern
Decryption Method	Reverse substitution used	Reverse rearrangement used
Example Techniques	Caesar Cipher, Playfair Cipher, Hill Cipher, Vigenère Cipher	Rail Fence Cipher, Columnar Transposition Cipher
Mathematical Operation	Uses substitution rules/modular arithmetic, matrix operations	Uses permutations and positioning
Vulnerability	Vulnerable to frequency analysis	Vulnerable to anagram analysis
Key Usage	Key decides substitution mapping	Key decides rearrangement order

Modern Usage	Basis of many modern encryption systems	Often combined with substitution for better security
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Cryptographic Key

A **cryptographic key** is a piece of information—typically a string of random characters, numbers, or bits—used by a cryptographic algorithm to transform readable data (**plaintext**) into an unreadable format (**ciphertext**), or vice versa. Without the key, the algorithm is just a useless machine. The key acts as the unique passcode that configures the mathematical operations of the algorithm for a specific session or user.

Kerckhoffs's Principle: In cryptography, the security of any encryption algorithm does not rely on keeping the mathematical formula a secret—it relies entirely on keeping the **Key** a secret. This principle is known as **Kerckhoffs's Principle**.

Key Size (Bit Length): refers to the length of the key measured in **bits**. It is the most critical factor determining how resistant an algorithm is to a brute-force attack (where an attacker tries every possible key combination).

- **Symmetric Key Sizes (e.g., AES):** Typically range from **128 bits** to **256 bits**. Because symmetric ciphers use direct mathematical scrambling, relatively short keys provide massive security.
- **Asymmetric Key Sizes (e.g., RSA):** Typically range from **2048 bits** to **4096 bits**. They require much larger key sizes because they rely on specific mathematical vulnerabilities (like the difficulty of factoring giant prime numbers).

Key Range (Key Space): **Key Range** is the total number of unique, possible keys that can be generated at a specific key size. If a key is n bits long, it is a binary string. Therefore, the key range is determined exponentially: 2^n

The Power of Exponential Growth: Every single bit you add to a key size **doubles** the size of the key range, making it twice as hard to crack. To see how drastically this scales, let's look at a comparative matrix:

Key Size (n)	Key Range (Total Possible Keys)	Security Status against Brute-Force
8 bits	2^8	Cracked in microseconds.
56 bits (Old DES standard)	2^{56}	Broken in 1998 in under 3 days; completely insecure today.
128 bits (Standard AES)	2^{128}	Secure. Would take supercomputers billions of years to brute-force.
256 bits (Military-grade AES)	2^{256}	Impenetrable. The number of combinations approaches the number of atoms in the observable universe. Quantum-resistant.

The Relationship Between Size, Range, and Security

A large key size guarantees a massive key range, but it only ensures security if the algorithm satisfies **Entropy (Randomness)**.

- **The Trap of Weak Key Generation:** If an algorithm uses a 256-bit key size (2^{256} range), but the software generating the key has a bug that only picks keys starting with twelve zeros, the *actual* effective key range shrinks drastically. Attackers can ignore the rest of the key space and break it instantly.
- **The Performance Trade-Off:** Larger keys require more CPU processing cycles and battery power to encrypt and decrypt data. Cryptographers always balance security requirements against computational efficiency (which is why AES-128 is still widely used over AES-256 for standard web traffic).

Types of Cryptography

1. Symmetric Key Cryptography

- Symmetric key cryptography is a type of encryption where the same key is used for both encryption and decryption.
- The sender and receiver must share the secret key securely before communication.
- Extremely fast and computationally efficient. It is designed to handle massive volumes of data (like streaming video, hard drive encryption, or database storage).
- **The Key Distribution Problem:** The biggest vulnerability of symmetric cryptography is *how* to safely share the secret key. If a hacker intercepts the key while A is sending it to b, the entire system is instantly compromised.
- Examples: AES, DES, Blowfish, RC4
- It is mainly used in: File encryption, Disk encryption, VPNs, Database security

Plaintext + Secret Key $\rightarrow$ Ciphertext

Ciphertext + Same Secret Key $\rightarrow$ Plaintext

2. Asymmetric Key Cryptography

- Asymmetric key cryptography uses two different keys: Public Key & Private Key
- The public key is used for encryption, and the private key is used for decryption.
- The public key can be shared openly, while the private key is kept secret.
- **Speed:** Highly complex and computationally heavy. It relies on advanced modular mathematics (like factoring massive prime numbers or calculating elliptic curves) and is much slower than symmetric encryption.
- **Solves the Key Problem:** No secret keys ever need to be exchanged over an insecure network. It is more secure for communication and key exchange
- Examples: RSA, ECC, Diffie-Hellman, ElGamal
- It is mainly used in: HTTPS/SSL, Digital signatures, Online banking, Secure email systems

Plaintext + Public Key $\rightarrow$ Ciphertext

Ciphertext + Private Key → Plaintext

Feature	Symmetric/ Private/ Secret/ Common Key Cryptography	Asymmetric Key Cryptography
Number of keys	One key (same for encryption & decryption)	Two keys (Public key + Private key)
Speed	Fast	Slow (computationally expensive)
Key distribution	Difficult and insecure over network	Easy (public key can be shared openly)
Scalability	Poor (n users need $n(n-1)/2$ keys)	Good (each user needs only one key pair)
Processing power	Low	High
Typical key size	Small (128, 256 bits)	Large (1024, 2048, 4096 bits)
Used for	Bulk data encryption	Key exchange, digital signatures, authentication
Examples	AES – Advanced Encryption Standard, DES – Data Encryption Standard, 3-DES	RSA – Rivest–Shamir–Adleman, Diffie–Hellman (DH) – Diffie–Hellman Key Exchange